Shopping Behaviour Insights: Data Analysis Report

An Exploratory and Statistical Analysis of Customer Purchasing Patterns

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Shopping Behaviour Insights

1. Data Exploration

Column Name	Data Type	Data Level	Description
Customer ID	Integer	Nominal	Unique identifier for CSV
Age	Integer	Ratio	The age for customer who have purchased from the shopping retailer
Gender	String	Nominal	Gender of customer
Item Purchased	String	Nominal	Item purchased from customer
Category	String	Nominal	The category the item is from
Purchase Amount (USD)	Float	Ratio	How much the item costs
Location	String	Nominal	Where the customer is located
Size	String	Ordinal	The size of the item purchased
Colour	String	Nominal	The colour of the item purchased
Season	String	Nominal	The season in which the customer purchased the item
Review Rating	Float	Interval	The rating left by the customer reviewing the item
Subscription	String	Nominal	The status of whether a customer has a subscription or not
Discount Applied	String	Nominal	Whether the customer used a discount when purchasing the item
Previous Purchases	Integer	Ratio	The number of previous purchased the customer has made
Payment Method	String	Nominal	The method of payment used by the customer
Frequency of Purchases	String	Ordinal	How often the customer purchases from the retailer

2. Insights

Age Groups

Firstly, uncovering key demographics such as age distribution shows that there is a low difference between overall between all age groups. The distribution of revenue throughout all age groups indicates that no group dominates the overall spending.

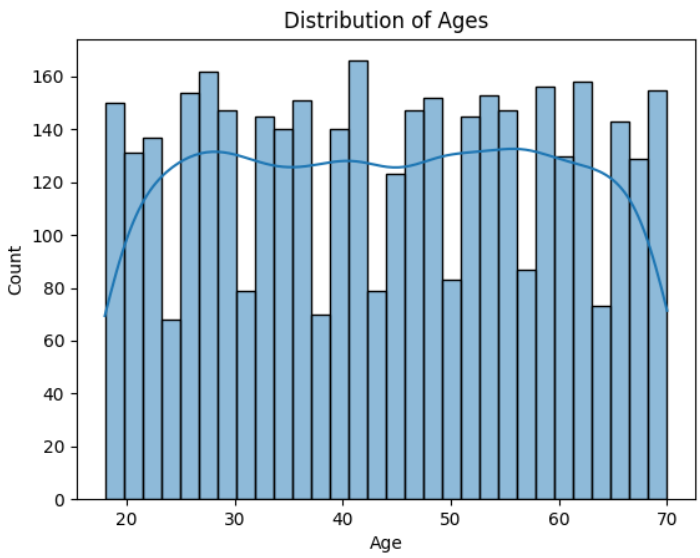

Figure 1 Age distribution plot

Revenue generated per Age Group:

Age Group	Revenue
18-25	$34,630
26-35	$44,342
36-45	$43,234
46-55	$45,619
56-65	$44,352
66+	$20,904

Note: 66+ group category has a maximum age of 70, therefore indicating that although it's the lowest age group in revenue, it's also the smallest age group.

The revenue generated per age group indicates that 18-25 spends the least, as 18-25 group comprises of 14.86% ($34630) of the total revenue.

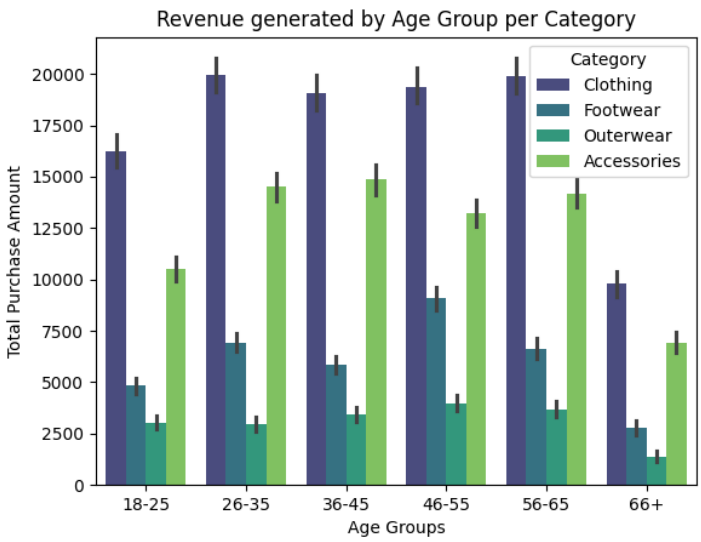

Figure 2. The revenue generated by each category for every age group

When reading the bar chart above, there is evidence that 18-25 is the lowest age group in all categories. Clothing is the highest performing category for 18-25, which returns a revenue of approximately $16250, whereas other age groups are generating between $19,000 - $20,000 in revenue.

To understand if age significantly affects average revenue per customer, a one-way ANOVA test was conducted, with a null hypothesis that age has no effect on revenue generated.

The test returned $F(5, 3894) = 0.60$, $p = 0.703$.The p value is greater than the 0.05 significance level (α), we fail to reject the null hypothesis that mean is equal through all age groups. Indicating that although the 18–25-year-old age group generates the least amount of revenue, there is no statistical evidence indicating to claim the reason for it being the lowest group is cause of age.

Conclusion: No statistically significant difference in average revenue by age group (ANOVA, p=0.703).

Subscribers

When looking at the data, based on subscribers and the correlation to previous purchases, there is a minimal variation for Subscribers and Non-Subscribers, as shown in the boxplot below:

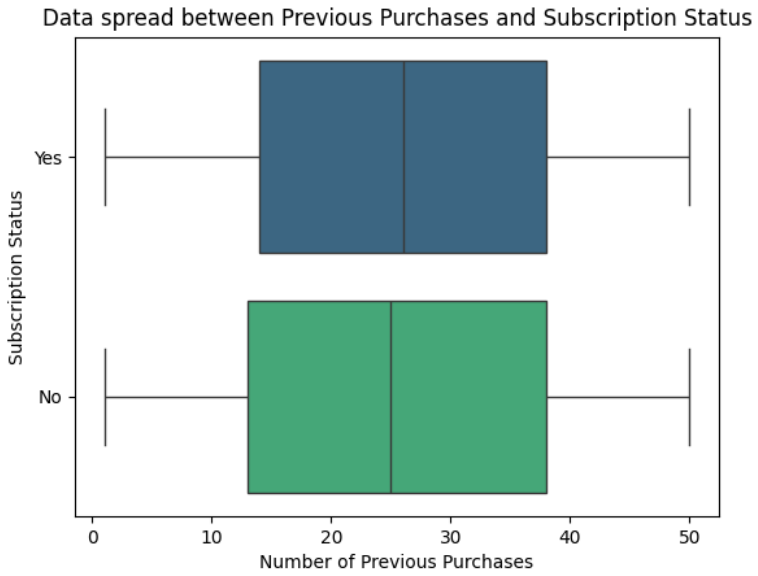

Figure 3. A box plot displaying the skewness of number of previous purchases in relation to subscription status

Using the Subscribed Status vs Previous Purchases, assists us in understanding if a subscribed user, orders more items, due to having a subscription. Analysing the illustration indicates that there is a minimal statistical significance between the averages of a subscribed user.

A t-test was performed to further investigate whether the variation between non-subscribers and subscribers was a great margin.

Null Hypothesis: Subscription Status has no effect on number of previous purchases.

The test returned, t-test = 1.953, and P-Value = 0.0509. With the averages of subscribers and non-subscribers being, 26.084 (Subscribed), 25.0804 (Not Subscribed). This means the difference between the average of subscription statuses of a customer is 1 purchase.

Based on the test conducted indicates that there is a low difference between the subscription status and number of previous purchases. However, the relationship

between Subscription Status and Previous Purchases does not reflect the revenue generated in accordance with the status of a customer's subscription

Subscription Status	Total Revenue
Yes	$62,645 (26.65%)
No	$170,436 (73.35%)

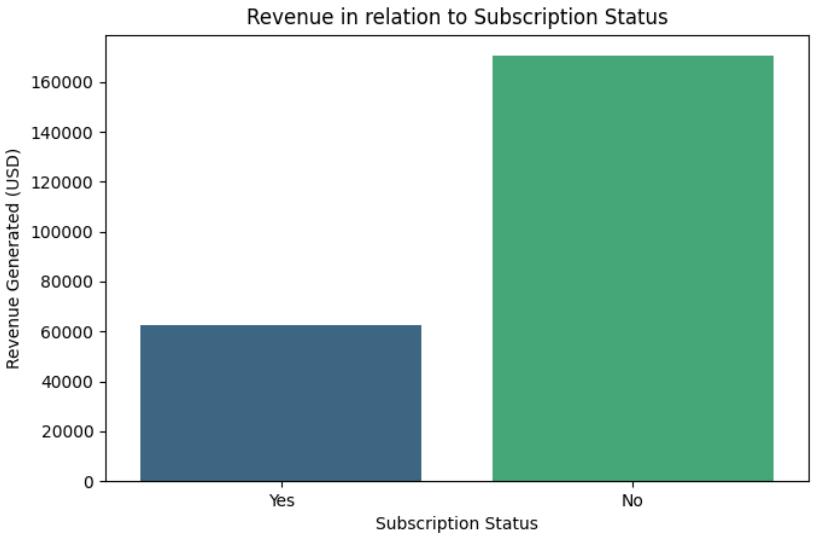

Figure 4. Revenue generated by subscribers and non-subscribers

As illustrated, the revenue generated by non-subscribers is significantly greater than customers who have subscribed. Subscribed customers have generated $62,645 which is only 26.87% of the total revenue, whereas non-subscribed customers have generated $170,436. However, this revenue does not include the subscription that subscribed users pay monthly.

Conclusion: The revenue generated does not have a strong correlation to subscription status of customers. This can be due to Subscriptions not having enough benefits for a customer, or customers don't feel urged enough to pay a monthly subscription however further analysis would need to be conducted to understand the underlying cause.

Discounts

When comparing the total number of customers who have applied discounts to the number of customers who have not applied discounts, you can see there is a frequency of customers who did not apply discounts, as indicated in the bar chart below:

Figure 5

The same relationship is shown when investigating the revenue generated in accordance with discount status.

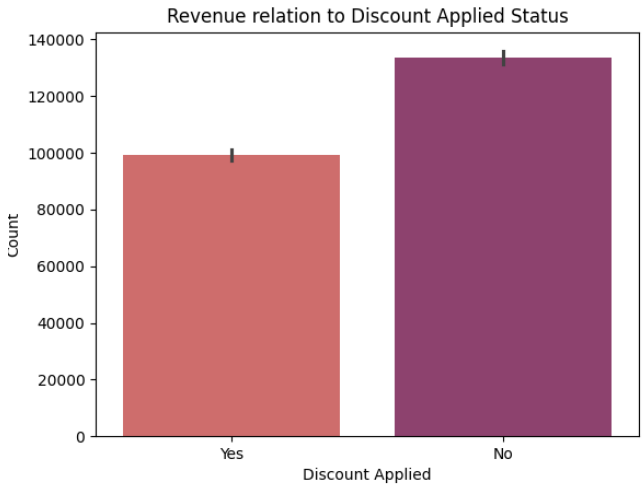

Figure 6 Revenue generated according to discount status

To identify whether the status of a customer's discount was an influence of how the customer reviewed the product. As a customer paying a lower price can influence them into interpreting the product as better or more valuable. As investigated, it was pleasant to identify that discount status did not influence how products where perceived, meaning that the overall review ratings for customers who applied discounts were 3.74 stars, and an average of 3.76 stars for customers who had not applied discounts. Both these discount statuses share a range of between 2.5 and 5.0 stars. Indicating that customers are honest along with that the products we sell are of good quality, based on

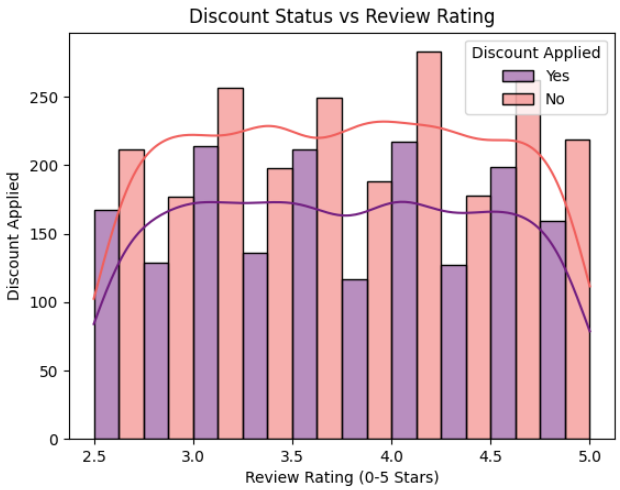

Figure 7 Review rating distributed according to discount status

the reviews it indicates that the customers appreciate and value our products. Based on the histogram shown below it further strengthens the conclusion that discount status has no effect on how a customer perceives a product.

Conclusion:

- **Review Ratings don't have a correlation to the discount status**
- **Products are of good quality as the minimum rating is between 2.5 and a max of 5 stars and an average of 3.75 stars**
- **Customers are honest about the products they purchase**

Customer Loyalty

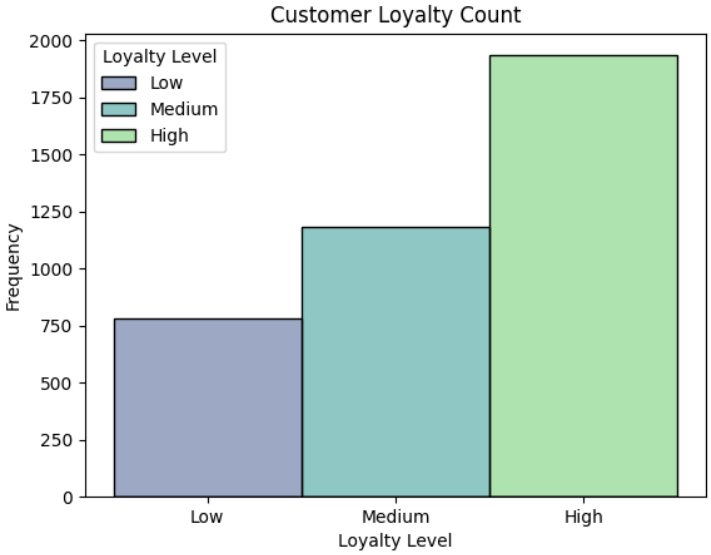

Figure 8: Loyalty Level count plot

The graph above displays the levels of loyalty based on the number of previous purchases. The reason for using previous purchases as a loyalty indicator, is that the number of previous purchases shows trust and an ongoing relationship with the retailer, therefore allowing this indicator to be used as a proxy.

As shown, there is a high quantity of loyal customers. This segment makes up 49.61% of the total customers. With customers with a low loyalty making up 20.10% of the total customers. This suggests that there is a current high volume of customers have repeated and continuous engagement with the retailer.

Revenue of each level:

- Low: $47,564 (20.41%)
- Medium: $69,555 (29.84%)
- High: $115,962 (49.75%)

The revenue generated by each sector shows a direct correlation between the previous purchases and how loyal a customer is, as high loyalty customers have generated 49.75% of all total revenue. Suggesting customers trust this retailer.

This acts as an indicator for customer retention, showing that customers are being kept close as they have purchased frequently, this also allows us to understand the churn as returning customers for a prolonged period of time are less likely to leave, however it is the responsibility of the retailer to have products that cater to the customer's needs.

Time Sensitive Insights

During different seasons throughout the year there is a shift in customer demands, as illustrated, clothing is the highest performing category compared to other categories. However, during Summer there is a massive drop in revenue, which allows it to match the second highest category, Accessories. This can be since there are not any products in the summer sector that appeal to customers

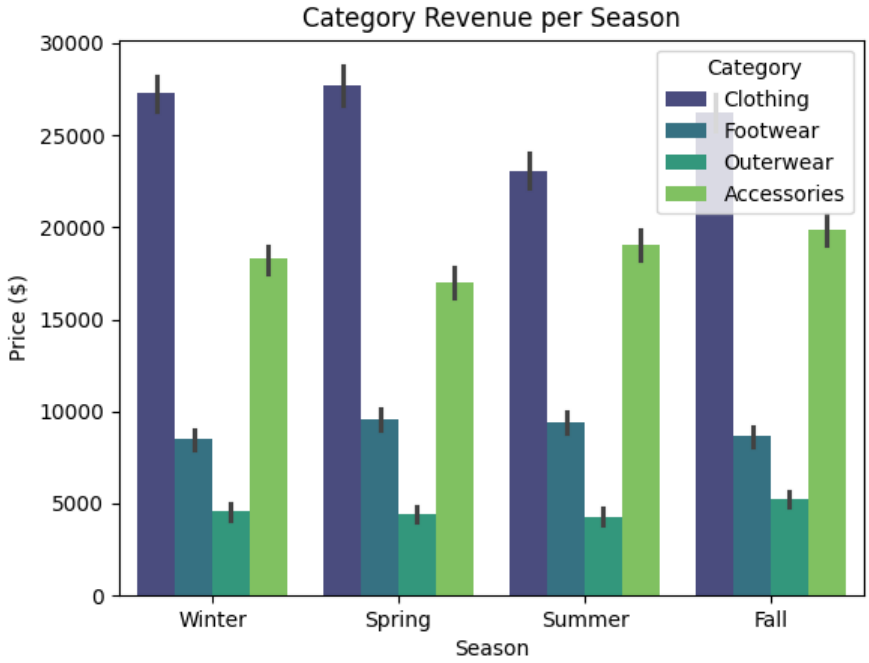

Figure 9. Total revenue generated by each category per season

Product Insights in relation to yearly performance

The first noticeable drop in revenue is in the clothing category during the summer, as the revenue generated that season from clothing was $23,078, compared to the performance in the previous season where Clothing generated $27,692, it is evident that there was a significant loss, the difference between the two seasons was 16.67% and $4,614.

When uncovering why other categories (e.g. Footwear and Outerwear) return a lower revenue compared to other categories, it is clear to see that it is due to the lack of products being offered in those categories, shown in the graphs below, it is illustrated that the outerwear only offers 2 products, suggesting that if a greater variety of products

were offered it could generate a higher return on revenue whilst still non-volatile, as with only 2 products it was the most consistent category throughout the year.

Secondly, footwear, offers only 4 products, while also like Outwear, being very consistent throughout the year, showing potential for another stream of higher and stable revenue throughout all seasons.

Lastly, Accessories, the second highest performing category has performed throughout the year, almost outperforming Clothing in Summer, whilst having 3 less products in its category. This indicates that if more products were added to Accessories, it to match and potentially outperform Clothing

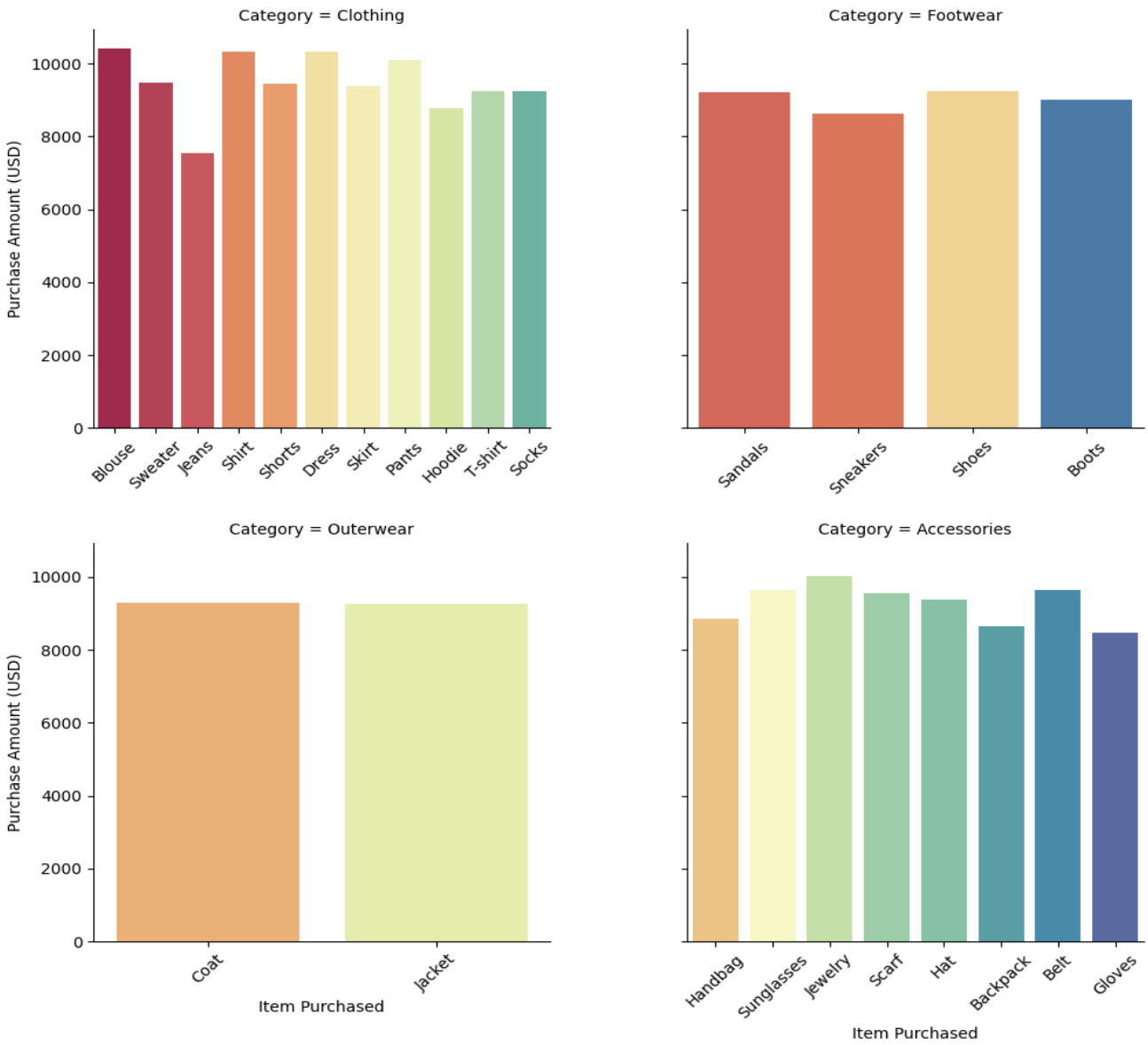

Figure 10 Bar plot illustrating the number of products sold per category

Location

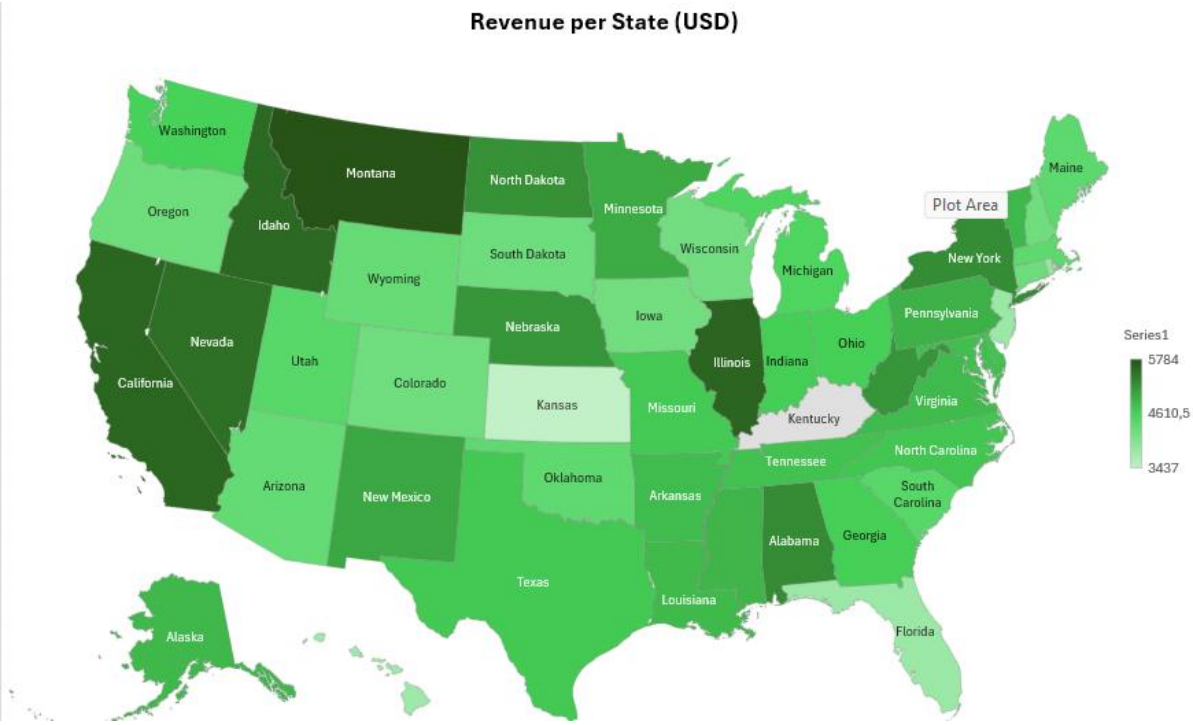

Figure 11. Heatmap of revenue generated per state

The graph above displays the total revenue generated that each state has generated. The state that delivered the lowest revenue is Kansas, with a revenue of $3,437. The state to deliver the highest revenue is Montana, generating a revenue of $5,784, this indicates that there is a $2,347 difference between the two states. The average revenue generated per state is $4,661.62, with 48 out of 50 states performing at and above average. The top 5 out of 50 states make up for 12% ($28,107) of the total revenue. Whilst the lowest performing state does not bring in the highest revenue, the positive is that 48 other states perform above the average.

Size

The sizes of items purchased displayed that the highest purchased size is Medium, whilst the least purchased size being Extra Large. With a difference of 1329 purchases between the 2 sizes. Small and Large were also purchased substantially by a lower amount compared to Medium. With Large having 698 purchases less than medium and small having 1092 purchases less.

Size	Count
S	663
M	1755
L	1053
XL	429

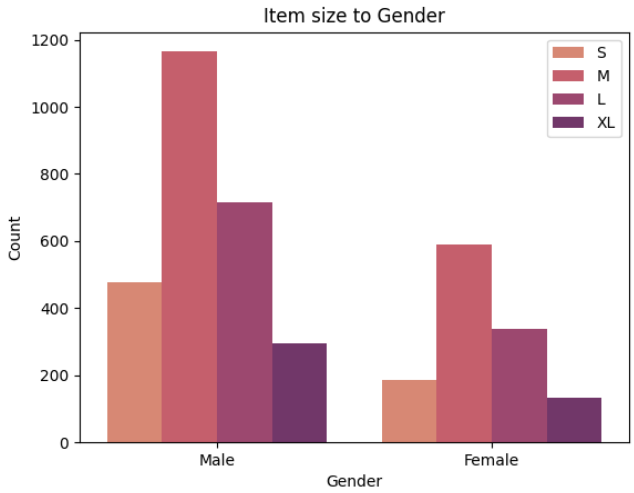

Figure 12 Products purchased by genders divided into sizes.

When this is paired with the gender category, which consists of 2652 Males, and 1248 Females, despite the greater number of males in the dataset, the demand for medium sized items is still relevant. With the same trend of size Small, Large and Extra Large being relevant.

Along with this, medium was the most purchased size in all states of America. Meaning that there is a country wide demand for medium across both genders.

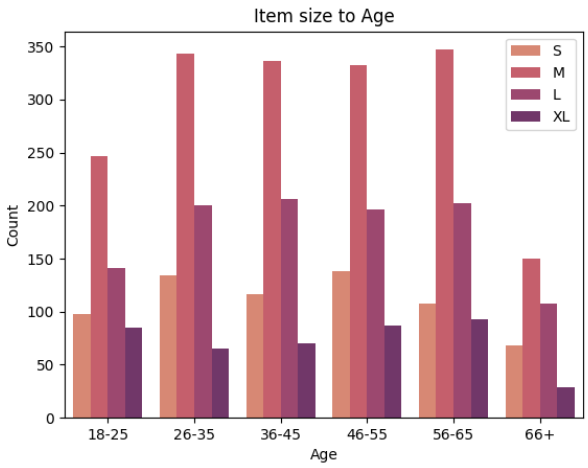

Figure 13 Sizes purchased across all age groups

The same trend appears to be relevant even when analysing the sizes purchased by each age group.

*Note: The maximum age in age group 66+ is 70, therefore being a smaller group.

As shown, the there is high purchase history for size medium, showing that regardless of gender and age, medium is still a popular size for products. Another pattern to uncover is that the purchase history for size Large indicates that less people from the age group 18-25 is less in comparison to the other age groups.

When conducting statistical tests to find the correlation between Size when coupled with Location and Gender. The following results were returned:

Gender vs Size

Chi-square: 6.615107840598041

p-value: 0.08523181331915769

The P-Value is greater than 0.05, therefore there is no evidence of any association with size in relation to gender.

Location vs Size

Chi-square: 144.63599866921743

p-value: 0.539697517214638

The P-Value returned is also greater than 0.05 meaning again that there is no evidence of any correlation between locations and the sizes purchased.

What does this mean?

That there is a stable and fair distribution of size across all demographics, and size independent of gender, age and location.

Payment Method

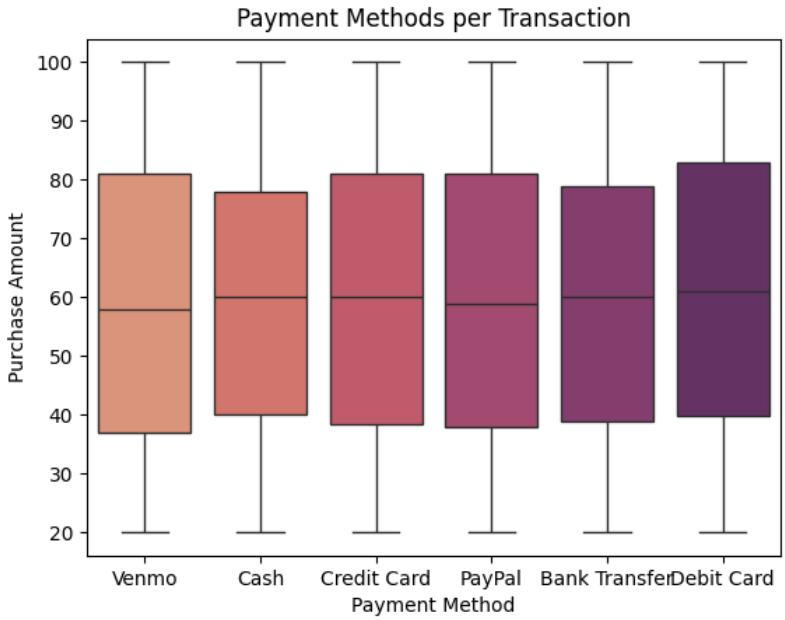

Figure 14 Box plot diagram illustrating the payment method used for various purchase amounts

Payment Method	Count
Bank Transfer	612
Venmo	634
Debit Card	636
Cash	670
Credit Card	671
PayPal	677

Shown above are the distributions for various payment methods used by customers when purchasing items on the online platform, as shown the median, minimum and maximum all have a small difference. Along with this is a table showing how much each payment method was used. As shown, bank transfers were the least, and PayPal was the highest used payment method. With a difference between the highest and lowest payment method being 65 purchases. This also displays an average of 644 purchases overall for each payment method. This according to the average, this shows that Bank Transfer, Venmo, and Debit Cards are used below the average amount of purchases. Whilst Cash, Credit Card and PayPal all performing within proximity of one another. As there is only a difference of only 7 purchases between Cash and PayPal.

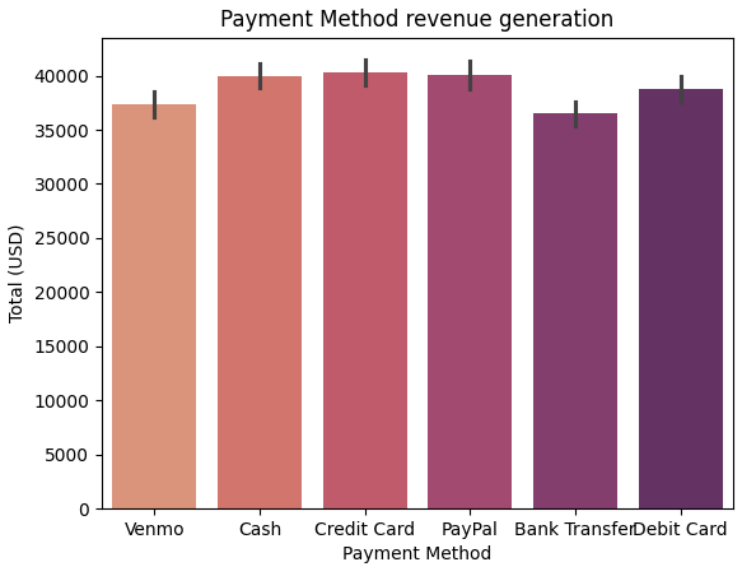

Figure 15: Revenue generated through all available payment methods offered

Payment Method	Total revenue
Bank Transfer	$36,544
Venmo	$37,374
Debit Card	$38,742
Cash	$40,002
PayPal	$40,109
Credit Card	$40,310

However, what correlates here is that most purchases used by customers, are done with a customer's own money, as credit cards only make up 17.21% of total purchases made. This indicates that customers prefer using their own money rather than lending from the bank on credit.

Key Takeaways

- **Age has no effect on revenue generated.**
- **Age has a little to no correlation with sizes of items**
- **Subscribers make up a small minority of customers**
- **No significant difference in average ratings between discounted and non-discounted purchases**
- **Items sold are of good quality**
- **More items need to be sold in the Footwear and Outerwear category**
- **High quantity of loyal customers**
- **Size has no relation to the gender of the customer**
- **Each state generates a similar revenue**
- **There are more male customers**

What is possible with this information?

- **Targeted ads**
- **Further analysis on value of subscription benefits**
- **Understand supply needs from customers**
- **Inventory Management**
- **Understand yearly revenue generators**
- **Further analysis into customer churn and retention**

Recommended Actions

- **Expand Footwear and Outerwear product line**
- **Fix revenue drops for clothing in summer**
- **Add more products to Accessories category**
- **Identify ways to add more value to subscription**
- **Create a VIP tier for high loyalty customers to increase retention**
- **Increase overall stock for medium sizes**
- **Reduce stock of XL products**
- **Focus on advertising non-age specific/targeted ads**
- **Advertise according to season and product**
- **Increase marketing in states that perform below average (e.g. Kansas)**